Discrete Electronics Project: Class AB Audio Amplifier

Design and Simulation Progress Report

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***Abstract*—This paper outlines the design and simulation progress of a discrete-component Class AB audio amplifier developed in collaboration with Analog One. The system is organized into three functional sections: preamplification, amplification, and power amplification. The proposed arrangement includes an audio input network, volume adjustment, filtering, and a high-impedance preamplifier output driving subsequent stages toward an $8\,\Omega$ load. Both the preamplifier stage (using a single $+24\,\mathrm{V}$ rail) and the Class AB power stage (using $\pm 24\,\mathrm{V}$ supplies) have been validated through LTspice and Multisim simulations. This report describes circuit topologies, component selections, DC operating points, time-domain transient responses using uncompressed audio streams, and AC sweep results.**

***Index Terms*—Class AB amplifier, preamplifier, discrete electronics, BJT, MOSFET, audio amplification, LTspice, simulation.**

I. Introduction

An audio amplifier must increase the level of an input signal while maintaining sufficient signal quality to drive a loudspeaker. Designing an amplifier using discrete electronic components requires attention to transistor biasing, coupling networks, gain, frequency response, and the electrical conditions of the output stage. In particular, a Class AB output stage aims to reduce the high quiescent dissipation associated with Class A operation while limiting distortion around the transition between output devices.

The Discrete Electronics Project, developed in collaboration with Analog One, focuses on a Class AB audio amplifier. Its architecture is divided into three functional sections: a preamplifier, an amplifier, and a power stage. Both the preamplification stage and the power stage have been evaluated in simulation as part of the overall development process. This report summarizes the proposed organization, explains component choices across stages, and presents the analytical and simulation methods used to assess the complete amplifier system.

II. Circuit Description

A. Preamplification Stage

The preamplification stage processes the incoming analog audio signal—tested using the raw waveform of "golden hour" by JVKE—and conditions it for the subsequent gain stage. Powered from a single $V_{CC} = +24\,\mathrm{V}$ supply, the circuit employs a 2N2222A BJT in a common-emitter (CE) topology configured to drive a nominal load of $R_L = 1\,\mathrm{M\Omega}$.

Voltage-divider biasing is established by $R_{B1} = 150\,\mathrm{k\Omega}$ (R_{10}) and $R_{B2} = 22\,\mathrm{k\Omega}$ (R_{11}), maintaining $V_B \approx 3.07\,\mathrm{V}$. The collector resistor is set to $R_C = 4.3\,\mathrm{k\Omega}$ (R_9). To adjust the AC voltage gain so that the output signal peak remains within the $\pm 1.0\,\mathrm{V_{peak}}$ range ($0\,\mathrm{dBFS}$ digital full-scale) without requiring external attenuator resistor dividers at the output, the emitter resistor R_E (R_8) was selected as $2.2\,\mathrm{k\Omega}$. The emitter bypass capacitor (C_E) was deliberately omitted to provide negative feedback, stabilizing midband gain and preserving linear performance across wide dynamic ranges:

$$A_v \approx \frac{R_C \parallel R_L}{R_E + r'_e} \approx \frac{4.3\,\mathrm{k\Omega} \parallel 1\,\mathrm{M\Omega}}{2.2\,\mathrm{k\Omega} + 5.6\,\Omega} \approx 1.94\,\mathrm{V/V} \quad (\approx +5.76\,\mathrm{dB}). \tag{1}$$

Coupling capacitors $C_{in} = 10\,\mu\mathrm{F}$ (C_4) and $C_{out} = 10\,\mu\mathrm{F}$ (C_2) isolate DC bias levels between stages. To ensure a completely flat frequency response across the entire human audible spectrum up to $22\,\mathrm{kHz}$, the input low-pass filtering capacitor C_1 was set to $100\,\mathrm{pF}$ (or $1\,\mathrm{nF}$). Combined with series input resistances, the high-frequency cutoff is pushed well into the megahertz range ($f_H \approx 1.59\,\mathrm{MHz}$ for $100\,\mathrm{pF}$), guaranteeing $0.0\,\mathrm{dB}$ attenuation at $22\,\mathrm{kHz}$.

B. Amplification Stage

The purpose of the voltage amplification stage is to increase the signal voltage before it reaches the power stage. It does not drive the loudspeaker directly. The final design target was approximately $22\,\mathrm{V}$ peak at the output, using a $2\,\mathrm{V_{RMS}}$ sinusoidal input and $\pm 24\,\mathrm{V}$ supplies. For a sine wave, this

output is approximately $44\,\mathrm{V}$ peak-to-peak or $15.56\,\mathrm{V_{RMS}}$, requiring a voltage gain of about $7.8\,\mathrm{V/V}$.

Early tests used two BS170 transistors and $\pm 30\,\mathrm{V}$ supplies, reaching approximately $13\,\mathrm{V_{RMS}}$. The design was then revised to meet the agreed $\pm 24\,\mathrm{V}$ supply and $22\,\mathrm{V}$ peak output requirements. The amplification stage uses three MOSFETs: two BS170 devices for voltage amplification and one IRF9510 to support the second stage. This stayed within the three-device allocation discussed for this section.

The two BS170 devices use a common-source arrangement. Each reverses the signal, so their two reversals give an overall output with the same polarity as the input in the middle of the audio range. The IRF9510 acts as an active load: it supplies current to the second BS170 instead of using only a resistor connected to the positive supply. It is not an additional common-drain amplifier or a loudspeaker driver.

The following component names refer to `SigAmpModule_LTspice.asc`. M1 and M2 are the BS170 devices, and M3 is the IRF9510. M1 uses a $330\,\mathrm{k\Omega}$ and $100\,\mathrm{k\Omega}$ gate divider, a $4.7\,\mathrm{k\Omega}$ drain resistor, and a $2.09\,\mathrm{k\Omega}$ source resistor. The source resistance was originally represented in Multisim by $1.27\,\mathrm{k\Omega}$ and $820\,\Omega$ in series. It has no bypass capacitor. A $4.7\,\mu\mathrm{F}$ capacitor connects the first stage to the second through a $53.3\,\mathrm{k\Omega}$ resistor.

M2 uses a $100\,\Omega$ source resistor connected to $-24\,\mathrm{V}$. Its gate connects to $-24\,\mathrm{V}$ through $22\,\mathrm{k\Omega}$ and to its drain through $220\,\mathrm{k\Omega}$. This feedback connection helps set its resting condition and control the gain. M3 connects to $+24\,\mathrm{V}$ through another $100\,\Omega$ resistor. Its gate divider uses $1\,\mathrm{k\Omega}$ toward $+24\,\mathrm{V}$ and $5.3\,\mathrm{k\Omega}$ toward ground. The drains of M2 and M3 join at the internal output. A second $4.7\,\mu\mathrm{F}$ capacitor couples this node to the external output.

TABLE I
FINAL COMPONENT VALUES IN THE AMPLIFICATION STAGE LTSPICE MODULE

Component	Value or device
M1, M2	BS170
M3	IRF9510
R1, R2	$330\,\mathrm{k\Omega}$, $100\,\mathrm{k\Omega}$
R3, R4	$4.7\,\mathrm{k\Omega}$, $2.09\,\mathrm{k\Omega}$
R5, R6	$53.3\,\mathrm{k\Omega}$, $22\,\mathrm{k\Omega}$
R7, R8	$100\,\Omega$, $220\,\mathrm{k\Omega}$
R9	$100\,\Omega$
R10, R11	$5.3\,\mathrm{k\Omega}$, $1\,\mathrm{k\Omega}$
C1, C2	$4.7\,\mu\mathrm{F}$ each

The standalone Multisim test used a $43\,\mathrm{k\Omega}$ output load as a resistive approximation of the following stage. The later LTspice test with the preamplifier used a $1\,\mathrm{M\Omega}$ output load. These are different test conditions. Connecting the actual power stage remains a separate check.

In Multisim, a nonpolar $1\,\mu\mathrm{F}$ capacitor coupled the standalone signal generator to the first BS170 gate. In the connected LTspice version, the preamplifier's output coupling capacitor provides the DC separation at this interface. The gain-stage input therefore retains its own resting gate voltage.

C. Class AB Power Stage

The power stage shown in Fig. 1 has already been tested in Multisim. Its role is to supply the current needed by the $8\,\Omega$ loudspeaker while following the audio signal received from the amplification stage. The $+24\,\mathrm{V}$ and $-24\,\mathrm{V}$ supplies provide room for the output signal to move in both directions around its resting level.

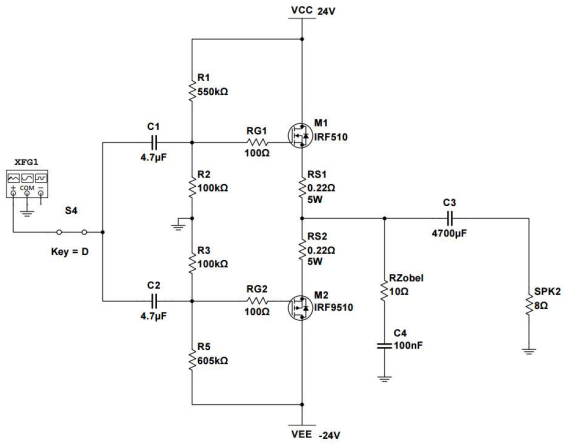


Fig. 1. Class AB power stage tested in simulation with an $8\,\Omega$ loudspeaker load and $\pm 24\,\mathrm{V}$ supply rails.

The IRF510 (M1) and IRF9510 (M2) were selected as a complementary pair of power transistors. Their opposite operating polarities allow M1 to mainly handle the positive portion of the signal and M2 the negative portion. These devices provide the current-handling function required at the output, and their packages allow heat sinks to be attached for physical implementation [1], [2].

Resistors $R_1 = 550\,\mathrm{k\Omega}$, $R_2 = 100\,\mathrm{k\Omega}$, $R_3 = 100\,\mathrm{k\Omega}$, and $R_5 = 605\,\mathrm{k\Omega}$ establish the resting conditions of the two transistors. The separate resistor ratios set their operating levels independently, while the high resistance values keep the current consumed by these networks small. Their purpose is to support a slight overlap between the activity of the devices, reducing distortion during the transition between positive and negative signal portions.

Capacitors C_1 and C_2, both $4.7\,\mu\mathrm{F}$, pass the audio signal while keeping the steady voltage of the preceding stage

separate from the transistor operating levels. Together with the input resistances, their capacitance keeps attenuation of low-frequency audio small.

The $100\,\Omega$ resistors R_{G1} and R_{G2} help suppress unwanted rapid oscillations at the transistor inputs. Their role is to support stable operation while allowing the audio signal to reach both devices [3].

The $0.22\,\Omega$ resistors R_{S1} and R_{S2} help stabilize the current through the output devices. Their low resistance keeps the signal voltage lost across them small. The $5\,\mathrm{W}$ rating provides capacity to dissipate the heat produced by the current flowing through these resistors.

At the output, the $4700\,\mu\mathrm{F}$ capacitor C_3 separates the loudspeaker from any steady voltage offset at the transistor output. This large capacitance was included because the $8\,\Omega$ load requires a substantial coupling capacitor to pass low-frequency audio with little attenuation.

The branch formed by $R_{\mathrm{Zobel}} = 10\,\Omega$ and $C_4 = 100\,\mathrm{nF}$ helps the stage operate with a more predictable load at high frequencies. This network was included because a real loudspeaker behaves differently as frequency changes. It provides a path for high-frequency current that helps reduce unwanted oscillations, while drawing little current at lower audio frequencies [3].

For the standalone simulation, XFG1 provides the test signal and S4 connects or disconnects it from the stage. In the complete amplifier, the preceding amplification stage supplies this input. The numerical results from the simulation will be presented separately from this description of the circuit and its component choices.

III. Simulation Description

A. Simulation Environment

Multisim and LTspice were used to simulate the audio amplifier, including the preamplification, amplification, and power stages. Both programs use mathematical models of the components to calculate voltages and currents throughout the circuit. This allows the design to be examined before physical assembly and helps identify how changes in one stage affect the others.

Multisim provides a graphical workspace with virtual instruments, such as signal generators and oscilloscopes, that support observation of the amplifier response. LTspice provides schematic-based simulation and waveform plots for examining voltages, currents, and frequency response. Both environments support the analysis of resting conditions, signal behavior over time, and operation across different frequencies.

A consistent comparison between the two simulators requires equivalent component values, supply voltages, input signals, and loudspeaker loads. The transistor models and simulation settings must also be considered when interpreting differences. For the complete amplifier, the analysis includes the connections between stages, since these connections influence the signal level and frequency response delivered to the loudspeaker.

B. DC Operating-Point Analysis

The DC operating-point analysis establishes the amplifier conditions without an applied audio signal. It calculates the steady voltages and currents that determine how the transistors are prepared to process the incoming signal. These conditions provide the starting point for interpreting the remaining simulations.

1) Preamplifier Stage: The DC operating-point simulation for the 2N2222A common-emitter preamplifier verified a stable Q-point under $V_{CC} = +24\,\mathrm{V}$. The base voltage set by the R_{10}–R_{11} resistor divider yields $V_B \approx 3.07\,\mathrm{V}$, establishing an emitter voltage $V_E \approx 2.37\,\mathrm{V}$ and an emitter current $I_E \approx 1.08\,\mathrm{mA}$ across $R_8 = 2.2\,\mathrm{k\Omega}$. Collector voltage rests at $V_C \approx 19.36\,\mathrm{V}$, providing symmetric dynamic margin above ground to accommodate dynamic signal swings without saturating or clipping the transistor.

2) Amplification Stage: The resting condition of M3 was adjusted first, with the Multisim input amplitude set to zero. Its initial gate divider used $10\,\mathrm{k\Omega}$ toward the positive supply. The resistor toward ground was adjusted until the shared drain voltage was close to zero and the current through the upper $100\,\Omega$ resistor was about $2\,\mathrm{mA}$. The selected divider initially used $10\,\mathrm{k\Omega}$ and $53\,\mathrm{k\Omega}$. Both values were later reduced by a factor of ten to improve the signal behavior without changing the divider ratio.

After the transfer to LTspice, a DC operating-point analysis was run from the main connected schematic. Internal nodes were saved explicitly so that the transistor resting voltages could be read. The results in Table II agreed closely with the earlier Multisim adjustment.

TABLE II
Measured DC Voltages in the Amplification Stage in LTspice

Measurement	Voltage
Positive / negative supply	$+24/-24\,\mathrm{V}$
M1 gate	$5.5814\,\mathrm{V}$
M1 drain	$15.9317\,\mathrm{V}$
M2 source	$-23.8063\,\mathrm{V}$
M3 source	$23.7963\,\mathrm{V}$
Shared M2–M3 drain	$0.177945\,\mathrm{V}$
External output after capacitor	Approximately $0\,\mathrm{V}$

The resistor voltage drops give approximately $2.037\,\mathrm{mA}$

through M3's source resistor and $1.937\,\mathrm{mA}$ through M2's source resistor. Their difference, about $0.100\,\mathrm{mA}$, flows through the feedback and gate-bias branch. M1's drain resistor gives a current of approximately $1.717\,\mathrm{mA}$. With its $2.09\,\mathrm{k\Omega}$ source resistance, the calculated M1 source voltage is approximately $3.59\,\mathrm{V}$. This last value is calculated from the measured drain voltage rather than recorded as a separate node measurement.

The shared drain rests close to zero instead of either supply rail. These readings support the intended resting condition of the amplification stage; they do not, by themselves, prove that every possible input signal will remain free of clipping.

3) Class AB Power Stage: The DC operating-point analysis examines the resting conditions of the power stage with the $+24\,\mathrm{V}$ and $-24\,\mathrm{V}$ supplies connected. The input source is configured without a DC offset, allowing the resistor networks to establish the operating levels of M1 and M2 independently of the incoming audio signal.

The analysis focuses on the transistor voltages and the current present when no audio signal is applied. These conditions determine how the devices remain ready to reproduce the positive and negative portions of the signal and how much power is consumed while the stage is idle.

The internal output voltage is examined before capacitor C_3, which separates the loudspeaker from any steady voltage at this point. The resting transistor voltages and currents are obtained from the operating-point results, separately from the measurements recorded during the audio signal test.

C. Transient Analysis

Transient analysis shows how the amplifier voltages and currents change over time when an audio signal is applied. Comparing the input and output waveforms allows examination of signal amplification, changes in waveform shape, and the response of the connected stages. The input amplitude, frequency, load, and simulation interval define the conditions of each test.

1) Preamplifier Stage: Time-domain transient simulations (`.tran 0 10s 1s`) were executed using a direct audio file stream (`golden_hour.wav`, AC = 1). Omitting the initial $1.0\,\mathrm{s}$ interval avoids transient DC coupling capacitor charging artifacts. Across the load $R_L = 1\,\mathrm{M\Omega}$ connected directly at node `out`, peak audio signal swings reach $\pm 1.0\,\mathrm{V_{peak}}$, corresponding to $0.707\,\mathrm{V_{RMS}}$ peak equivalent. Due to the crest factor and dynamic range of the audio material, the integrated RMS voltage over the $9\,\mathrm{s}$ window measures approximately $290\,\mathrm{mV_{RMS}}$. Exporting the audio directly via LTspice directive `.wave "amp_golden.wav" 16 48k V(out)` yields a clean, unclipped audio file matching standard digital full-scale limits.

2) Amplification Stage: Sine-Wave Tests: The standalone stage was tested in Multisim at $1\,\mathrm{kHz}$, increasing the input gradually. With the interstage resistor adjusted to $53.3\,\mathrm{k\Omega}$, an input of $0.2\,\mathrm{V_{RMS}}$ produced approximately $1.56\,\mathrm{V_{RMS}}$, and $1\,\mathrm{V_{RMS}}$ produced approximately $7.80\,\mathrm{V_{RMS}}$. Both tests supported the intended gain of about $7.8\,\mathrm{V/V}$.

At $2\,\mathrm{V_{RMS}}$, the output initially reached approximately $15.5\,\mathrm{V_{RMS}}$, but a visible bend appeared near the positive peak. The waveform at M1's drain appeared smoother, so the investigation focused on the M2–M3 section. Measurements across M3's source resistor showed a current variation coinciding with the bend. The M3 gate voltage also moved by roughly $170\,\mathrm{mV}$ peak-to-peak.

A trial $1\,\mu\mathrm{F}$ capacitor across the upper gate-divider resistor did not improve the observed waveform and was removed. Reducing the simulation maximum time step to $1\,\mu\mathrm{s}$ also did not remove the bend. The gate divider was then changed from $10\,\mathrm{k\Omega}$ and $53\,\mathrm{k\Omega}$ to $1\,\mathrm{k\Omega}$ and $5.3\,\mathrm{k\Omega}$. This kept the same nominal gate voltage while making it less sensitive to signal-related changes. The gate variation decreased and the pronounced bend was no longer visible in the output trace.

With the revised divider, a $2\,\mathrm{V_{RMS}}$, $1\,\mathrm{kHz}$ input produced measured output peaks of $+21.965\,\mathrm{V}$ and $-21.992\,\mathrm{V}$, or $43.957\,\mathrm{V}$ peak-to-peak. The reported RMS output was $15.6\,\mathrm{V_{RMS}}$. This met the approximate $22\,\mathrm{V}$ peak target with the $43\,\mathrm{k\Omega}$ test load. The waveform was assessed visually; no numerical total harmonic distortion result was obtained.

3) Amplification Stage Connected to the Preamplifier: The two blocks were then connected in LTspice using a $20\,\mathrm{s}$ transient simulation and the supplied `golden_hour.wav` file. This integration record belongs to the version supplied for that test, whose preamplifier used a $510\,\Omega$ emitter resistor and a $1\,\mathrm{nF}$ input filter capacitor. It is distinct from the $2.2\,\mathrm{k\Omega}$ preamplifier version described elsewhere in this report. The following observations do not claim to validate that later version. The power-stage block was not included in this connected test.

Before testing, the LTspice prefix of M2 was corrected to `X`, matching the supplied BS170 subcircuit model. The input and output waveforms were then compared. The connection between the blocks had a resting voltage of approximately $5.5814\,\mathrm{V}$, set by M1's gate divider. This value was subtracted only in the plot to display the changing audio signal around zero; the circuit itself was not modified by that plotting operation.

The original input to the amplification stage contained peaks of approximately $4\,\mathrm{V}$ relative to its resting level. At a gain

of about 7.8, these peaks would require roughly 31 V at the output. The ±24 V supplies could not support that voltage, and the plotted output developed flat negative peaks close to −24 V. Comparing the measured output with 7.8 times the input signal made this clipping visible. A music signal can have large brief peaks, so its RMS level alone does not establish whether it will fit within the amplifier's available output range.

For a controlled integration test, the music amplitude was reduced by half before the preamplifier using a behavioral source:

```
Bscale AUX 0 V=0.5*V(AUX_RAW)
```

This was a simulation-level input adjustment, not a change to the resistors or gain of the amplification stage. In the inspected interval from 16 to 16.02 s, the flat peaks disappeared and the output closely followed the scaled input. The full-song overview showed peaks mostly within approximately ±16 V, with a larger initial excursion that was not separately characterized. A physical implementation still needs an appropriate input-level or volume adjustment.

4) Audio Export During the Integration Test: A separate issue affected the exported WAV file. Directly exporting the amplifier output can clip the audio file because LTspice's WAV output uses a full-scale range of −1 to +1 V. The amplifier output was much larger than this range. A separate behavioral source therefore created a reduced copy for export:

```
Bexport AUDIO_EXPORT 0 V=V(VSPK)/30
```

The export command was changed to use that copy at the original file's 48 kHz sample rate:

```
.wave "AmpOutputMusic_Normalizado.wav"
16 48000 V(AUDIO_EXPORT)
```

The two displayed lines above form one directive in LTspice. Dividing this exported copy by 30 does not reduce the voltage delivered by the simulated amplifier. Playback of the new file was reported to sound clean. This listening check supports the improvement but is not a numerical distortion measurement.

5) Class AB Power Stage: The proposed input to the power stage is represented by a sinusoidal test signal at 10 kHz, with a peak amplitude of 22 V, equivalent to 44 V peak-to-peak. This source represents the drive signal that the preceding amplification stage is intended to provide.

With the 8 Ω load connected, the simulation recorded an output voltage of approximately 38.1 V peak-to-peak, equivalent to an effective value of 13.4 V_{RMS}. The corresponding load current was approximately 1.68 A_{RMS}, and the reported average power delivered to the load was 22.5 W under this test condition.

The output voltage is somewhat lower than the input voltage, consistent with the role of this stage: following the incoming signal while supplying the current required by the loudspeaker. M1 mainly handles the positive portion of the signal, while M2 mainly handles the negative portion.

The same test reported an average voltage component of approximately 1.41 mV across the load. This small value is consistent with capacitor C_3 separating the loudspeaker from the steady voltage at the transistor output.

The input and output waveforms are also examined for flattened peaks and irregularities near the transition between the two transistors. This waveform review complements the reported voltage, current, and power values when assessing signal quality.

D. AC Sweep and Frequency Response

The AC sweep examines how the amplifier responds as the input frequency changes. The resulting plots show the voltage gain and the relative phase between the input and output signals. This analysis identifies the frequency range over which the amplifier maintains a reasonably consistent response and the regions where the signal begins to decrease.

1) Preamplifier Stage: An AC sweep (`.ac dec 100 1 1e6`) confirmed a constant voltage gain across the entire audible spectrum (20 Hz to 22 kHz). Selecting $C_1 = 100$ pF successfully eliminated premature high-frequency attenuation near 10 kHz, extending the upper −3 dB cutoff frequency f_H past 1.0 MHz and maintaining 0.0 dB relative roll-off at 22 kHz.

2) Frequency Check During Amplification Stage Integration: A separate copy of the main schematic was used for an AC sweep with the directive `.ac dec 100 1 1Meg`. The source had an AC amplitude of 1, and the input scaling factor of 0.5 remained in place. The output load was 1 MΩ. Table III records the measured output magnitude, referenced to the original AC source before the scaling block.

TABLE III
CONNECTED PREAMPLIFIER AND GAIN-STAGE AC RESULTS

Frequency	Magnitude	Relative change
20 Hz	24.168 dB	−0.009 dB
1.002 kHz	24.177 dB	Reference
20 kHz	23.387 dB	−0.790 dB

The response was nearly constant through most of the audio range. At 20 kHz, the magnitude was about 0.79 dB below the

middle-frequency reference, corresponding to approximately 9% less amplitude. No pronounced gain peak was visible in the plotted audio band. This small decrease does not explain the severe flat peaks seen in the earlier music test, which were associated with excessive signal amplitude.

These results describe the connected preamplifier and voltage amplifier, including the 0.5 input factor, in the integration version identified above. They are not the isolated gain of the amplification stage and do not include the power stage. An isolated gain-stage response would require plotting its output divided by its own input. Also, the AC sweep examines small signals around the resting conditions; it does not establish the maximum unclipped output at every frequency.

3) Class AB Power Stage: The frequency-response analysis examines how consistently the power stage transfers the audio signal as frequency changes. The voltage across the loudspeaker is compared with the input voltage before capacitors C_1 and C_2, allowing the response of this stage to be assessed separately from the preceding amplification stages.

The signal test at $10\,\mathrm{kHz}$ provides a reference condition, with an input of $44\,\mathrm{V}$ peak-to-peak and a recorded output of approximately $38.1\,\mathrm{V}$ peak-to-peak. These measurements describe the response during that specific signal test. The AC sweep extends the assessment across a range of frequencies using small changes around the resting operating conditions.

At low frequencies, capacitors C_1, C_2, and C_3 influence how much of the signal reaches the loudspeaker. At higher frequencies, the transistor characteristics, input resistors, and connections between stages influence the output level. The complete frequency-response plot is used to identify the lower and upper limits of the useful operating range.

The power-stage response is considered together with the filter and the preceding stages when assessing the audio range of the complete amplifier. The reported $22.5\,\mathrm{W}$ output belongs to the signal test at $10\,\mathrm{kHz}$; power delivery at other frequencies is assessed through the corresponding time-based tests.

IV. Results to Be Documented

The preamplifier and power stages have been validated in simulation. Table IV records the performance parameters of the preamplifier stage, while Table V outlines the current documentation status of the power stage.

TABLE IV
PREAMPLIFIER STAGE SIMULATION RESULTS

Parameter	Target / Value	Status
Preamplifier Topology	2N2222A CE (No C_E)	Validated
Supply Voltage (V_{CC})	$+24\,\mathrm{V}$ DC	Verified
Midband Voltage Gain (A_v)	$+5.76\,\mathrm{dB}$ ($1.94\,\mathrm{V/V}$)	Verified
Peak Output Voltage	$\pm 1.0\,\mathrm{V_{peak}}$ ($0.707\,\mathrm{V_{RMS}}$ eq.)	Verified
Lower Cutoff Frequency (f_L)	$< 10\,\mathrm{Hz}$	Verified
Upper Cutoff Frequency (f_H)	$> 1.0\,\mathrm{MHz}$ ($0\,\mathrm{dB}$ at $22\,\mathrm{kHz}$)	Verified
Load Impedance (R_L)	$1\,\mathrm{M\Omega}$	Verified

TABLE V
CLASS AB POWER STAGE DOCUMENTATION STATUS

Parameter	Status
Power-stage simulation	Tested
DC bias conditions	To be documented
Midband voltage gain (dB)	To be documented
Lower cutoff frequency (Hz)	To be documented
Upper cutoff frequency (Hz)	To be documented
Output waveform and distortion	To be documented
Output voltage across $8\,\Omega$	To be documented

A. Amplification Stage Results and Remaining Checks

The amplification stage design progressed from the initial two-transistor stage to a three-MOSFET circuit operating from $\pm 24\,\mathrm{V}$. The standalone $1\,\mathrm{kHz}$ test reached the requested approximate $22\,\mathrm{V}$ output peak with a $2\,\mathrm{V_{RMS}}$ input and a $43\,\mathrm{k\Omega}$ load. The subsequent LTspice operating-point readings closely matched the Multisim resting conditions.

TABLE VI
SUMMARY OF AMPLIFICATION STAGE TESTS

Test or quantity	Recorded result
Standalone input	$2\,\mathrm{V_{RMS}}$, $1\,\mathrm{kHz}$
Supply rails	$\pm 24\,\mathrm{V}$
Standalone output load	$43\,\mathrm{k\Omega}$
Output peaks	$+21.965 / -21.992\,\mathrm{V}$
Output peak-to-peak	$43.957\,\mathrm{V}$
Reported output RMS	$15.6\,\mathrm{V_{RMS}}$
Approximate voltage gain	$7.8\,\mathrm{V/V}$
LTspice integration load	$1\,\mathrm{M\Omega}$
Music test correction	Input multiplied by 0.5
Audio-file correction	Export copy divided by 30

The integration tests identified two different sources of distortion: excessive drive into the amplifier and excessive amplitude when writing the WAV file. Lowering the test input addressed the first issue; scaling only the exported copy addressed the second. Neither adjustment changed the voltage gain resistors of the amplification stage.

The remaining checks for this section are operation with the actual power-stage input connected, large-signal tests near

the ends of the audio band, and physical measurements. A numerical distortion result and a full characterization of the initial music transient were not obtained. The $1\,\mathrm{M}\Omega$ integration load must not be interpreted as a loudspeaker or as proof of operation with the final power-stage load. These limits keep the documented results tied to the conditions actually tested.

V. Conclusions and Next Steps

The current design organizes the Class AB audio amplifier into preamplification, amplification, and power stages. Both the preamplifier and power stage have been validated in simulation, supporting linear dynamic range expansion, current delivery, separation of steady voltages between stages, and stable signal handling. The next steps are to integrate the intermediate amplification stage, evaluate inter-stage loading effects, and prepare for physical circuit assembly. Conclusions about total gain, delivered power, distortion, bandwidth, and thermal management will be supported by integrated simulation records and physical measurements.

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